



Team : Go Paloks

Livestock Management System

CSE 4408: System Analysis and Design Lab

Presentation 5

June 03, 2026

Team Members

- Sadnan - 230041119
- Tawfiq - 230041129
- Irtisham - 230041131
- Abtahi - 230041135
- Arkam - 230041157

Project Overview

Key Issues

- Manual record keeping
- Disconnected subsystems
- Poor resource tracking
- Inefficient reporting
- Delayed decision making
- No centralized monitoring

Core Features

- Integrated system
- Automated finance record
- Optimized resource planning
- Data driven decisions
- Real time monitoring
- Animal health tracking

Information Gathering

Need to Understand

- Actual user needs
- Existing workflows
- Pain points
- Reporting requirements
- System expectations
- User priorities

Used Methods

- Stakeholder Interviews
- Questionnaires
- Storytelling
- Requirement Elicitation Analysis
- JAD Feasibility Assessment

Objectives

Farm Owner

- Business goals
- Reporting needs
- Profitability concerns
- Expansion planning

Farm Manager

- Operational workflow
- Resource allocation
- Inventory management
- Scheduling issues

Workers

- Daily activities
- Record keeping difficulties
- Mobile usability requirements

Veterinarian

- Health monitoring process
- Vaccination records
- Disease management

Interview Structure

Pyramid Structure

- Specific -> General
- Closed -> Open Ended

Useful when:

- Warmup needed
- Interviewee seems hesitant

Maybe useful for:

- Farm workers, Managers

Funnel Structure

- General -> Specific
- Open Ended -> Closed

Useful when:

- Discovering interviewee's priorities
- Interviewee is knowledgeable

Maybe useful for:

- Farm owner, Veterinarians

Diamond Structure

- Specific -> General -> Specific
- Closed -> Open Ended -> Closed

Core Advantage:

- Combines strengths of the previous two structures

Questionnaire Design

Objective:

One to one conversation with a larger user group regarding:

- Current system challenges
- Technology readiness
- Desired features
- User satisfaction

Target Respondents:

- Farm Owner
- Veterinarians
- Farm Managers
- Workers

Interview Guide (Example : Manager)

Open Ended Questions

- Describe a typical workday.
- What tasks consume most of your time?
- What problems occur most frequently?

Closed Questions

- Do you currently maintain digital records?
- Are inventory records updated daily?
- How do you maintain worker shift scheduling?

Probe Questions

- Why does that issue occur?
- How often does it happen?
- What would improve this process?

Common issues among all stakeholders

1. Poor record keeping

- Manual record keeping
- Difficult tracking of all issues
- Mistaken Animal health history

2. Lacking of Real time data

- No live updates on anything
- Delayed decision making

3. Communication Gaps

- Weak connection amongst the different parts of the farm.

4. Manual & time-consuming processes

- Lack of use of modern technology
- Repetitive & redundant processes

User Specific Issues

1. Farm Owners

- Profit visibility
- Investment Decision Uncertainty
- Long term planning difficulty

2. Veterinarians

- Incomplete Medical History
- Medicinal misuse
- Late case reporting

3. Farm Managers

- Operational Overload
- Lack of Central Overview of the system

4. Farm Workers

- Lack of Clear Instructions
- Unclear technical skills

Storytelling Analysis (Example : Worker)

Example Story

"I record feed usage and animal health information in a notebook every day. During busy periods, some records are delayed or missed. When the manager requests information, finding the correct records takes time and sometimes important details are overlooked."

Problems Identified

- Manual record keeping
- Missing or delayed entries
- Time consuming record retrieval
- Inaccurate information sharing

Requirements Extracted

- Digital record management
- Mobile friendly data entry
- Centralized database
- Real time reporting
- Automated reminders

Requirement Elicitation

Operational Requirements

- Centralized livestock records
- Daily feed consumption tracking
- Inventory monitoring

Health Management Requirements

- Vaccination scheduling
- Treatment history maintenance
- Disease outbreak alerts

Reporting Requirements

- Financial summaries
- Livestock performance reports
- Inventory status reports

User Requirements

- Role-based access control
- Mobile accessibility
- Easy-to-use interface
- Local language interface

JAD Workshop Feasibility

Advantages

- Faster requirement gathering
- Better stakeholder consensus
- Reduced misunderstandings

Challenges

- Busy schedules
- Multiple farm locations
- Difficulty assembling all stakeholders

Conclusion

A small scale JAD session involving:

- Owner
- Manager
- Veterinarian

would be beneficial during detailed analysis.

Requirements Summary

High Priority

- Inventory Management
- Automated Report Generation
- Real-Time Dashboard
- Health & Vaccination Tracking
- Financial Analytics

Medium Priority

- Production Performance Analysis
- Supplier Management
- Data Visualization Dashboard
- Notification & Alert Customization
- Mobile Friendly Data Entry

Low Priority

- IoT Sensor Integration
- Marketplace Integration
- Predictive Disease Detection
- AI based Recommendations
- Third Party System Integration



Thank You

We are ready for your valuable feedback!